

CHAPTER 9

Control of Prolactin Production by Estrogen

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I. INTRODUCTION

The stimulation of prolactin by estrogen was first recognized almost 50 years ago (reviewed in Turner, 1977), and the means by which this stimulation occurs has been investigated by many laboratories. Estrogen affects almost every conceivable aspect of prolactin production; this chapter reviews

the processes that are changed and what we know about the mechanisms by which estrogen causes these changes.

II. ESTROGEN STIMULATES THE GROWTH OF PROLACTIN-PRODUCING CELLS

A. IDENTIFICATION OF PROLACTIN-PRODUCING CELLS

Prolactin cells have been distinguished from other cell types in the pituitary gland by three different methods. The secretory granules in which prolactin is stored can be stained with acidic dyes, and this technique was the earliest used to identify prolactin cells. Later, identification depended on morphological characteristics detectable with the electron microscope; prolactin cells are the only cells in the gland that may contain granules as large as 500–900 nm in diameter (Farquhar, 1977). More recently, the ability to visualize prolactin in the cells by immunocytochemical techniques has been used to identify prolactin cells. Several laboratories found using this technique that the pituitary gland contains prolactin cells that have only small granules in addition to the cells with the large secretory granules (Baker, 1970; Nogami and Yoshimura, 1980; Sato, 1980).

Use of the first technique, staining the granules with acidic dyes, missed cells that had few granules. Chromophobe cells that did not stain with acidic or basic dyes were often classified as cells not producing hormones, but subsequent studies showed that some of them were producing large amounts of prolactin but storing little. Identification of prolactin cells by granule size alone means that only a subpopulation of prolactin cells may have been examined. Immunocytochemical techniques can stain prolactin in the rough endoplasmic reticulum and Golgi apparatus as well as prolactin in granules, which means that cells that are synthesizing but not storing prolactin may be detected (Tougard *et al.*, 1980). This method, therefore, is the most extensive way of examining prolactin cells, and the emphasis in this discussion will be on results obtained using immunostains when possible.

B. ESTROGEN INCREASES THE NUMBER OF PROLACTIN CELLS

Gersten and Baker (1970) demonstrated that estrogen pellets implanted in the anterior pituitary gland of ovariectomized rats caused hyperplasia of prolactin cells, revealed by immunological staining. Corenblum and associates (1980) did a quantitative study of the effects of daily injections of es-

tradiol on the number of prolactin cells present in the glands of male rats with and without treatment with estradiol, detecting the cells by immunostaining. They found the percentage of cells that stained for prolactin increased from 15% in untreated male rats to 50% after 22 days of estradiol treatment. There are two experiments that indicate that the increase in the number of prolactin cells is a direct effect of estradiol on the anterior pituitary gland. The first experiment was performed in intact animals. In the study by Gersten and Baker, estrogen increased the number of prolactin cells only in the lobe in which the pellet was planted, but not in the other lobe. In addition, the estrogen effects were localized to particular parts of that lobe, only occurring in areas where the blood flow had first passed by the pellet and not occurring upstream of the pellet. The second experiment was performed in primary cell cultures of anterior pituitary glands. Lieberman *et al.* (1982) demonstrated that a 5-day incubation with estradiol increased the number of prolactin cells to 130% of the number in control cultures. Both experiments indicate that estrogen can act directly at the level of the pituitary gland to stimulate cell number, but do not indicate a direct action on the prolactin cells themselves, since the cells in the gland are heterogeneous.

The mitotic activity of the pituitary gland changes during the estrous cycle (Hunt, 1943). The changes may be caused by estrogen, because estrogen increases mitotic activity in the pituitary gland 2 days after injections in female and male rats (Hunt, 1947; Lloyd *et al.*, 1975). Treatment with estrogen also increases the ability of isolated pituitary glands or pieces of glands to incorporate [^3H]thymidine into DNA, which is interpreted as an increase in DNA synthesis. There may be problems with using [^3H]thymidine as a measure of DNA synthesis. Changes in the amount of incorporation may reflect changes in the specific activity of the precursor pools, not changes in the rate of synthesis, and incorporation may reflect repair of existing DNA, not synthesis of new DNA. One of the most thorough studies of [^3H]thymidine incorporation is that of Wiklund and Gorski (1982). They demonstrated that *in vivo* treatment of rats with estradiol did not change the specific activity or the size of the pool of thymidine triphosphate during a subsequent *in vitro* incubation of the glands with [^3H]thymidine. These workers also showed that increased incorporation occurred in isolated nuclei and that 80–90% of this activity had the characteristics expected if the enzyme involved was the DNA polymerase thought to be responsible for new DNA synthesis rather than the polymerase involved in repair. These data indicate that stimulation of [^3H]thymidine incorporation into the pituitary gland is a good reflection of DNA synthesis.

The cells that take up [^3H]thymidine in the pituitary gland include prolactin cells; 3% of the prolactin cells in male rats incorporated [^3H]thymidine *in*

vivo, determined by a combination of immunoperoxidase staining and autoradiography (Corenblum *et al.*, 1980). Mitotic cells that stain immunologically for prolactin have been seen (Sato, 1980), indicating that at least some of the new prolactin cells come from previously existing cells; whether others come from stem cells that differentiate into prolactin cells after dividing is not known.

III. ESTROGEN AFFECTS THE MORPHOLOGY OF PROLACTIN-PRODUCING CELLS

Estrogen increases the size of the prolactin cells in addition to increasing prolactin cell number (Gersten and Baker, 1970). Many laboratories have studied the changes in the morphology of the cell by electron microscopy. These studies are reviewed by Farquhar (1977); the changes include increases in the size of the Golgi apparatus and increases in the amount of rough endoplasmic reticulum, which is often found arrayed in long rows parallel to the plasma membrane or in whorled formations. A more recent study combined the techniques of immunoperoxidase staining with electron microscopy to examine the pituitary glands of mature female rats before ovariectomy, 4 weeks after ovariectomy, and after treating the ovariectomized rats with estrogen for 4 days (Osamura *et al.*, 1982). In normal female rats, they found staining for prolactin in the rough endoplasmic reticulum, in the Golgi saccules, and in membrane-bound secretory granules. The cells contained the large secretory granules (about 600 nm in diameter) commonly associated with prolactin cells. After ovariectomy, the size of the cells that stained for prolactin was markedly decreased, and both the amount of the rough endoplasmic reticulum and the size of the Golgi apparatus were reduced. The cytoplasm was filled with small secretory granules (120–190 nm) that stained for prolactin. When estrogen was given for 4 days, the most prominent feature that stained positively for prolactin was the rough endoplasmic reticulum that was found in the characteristic large whorls or in rows parallel to and close to the plasma membrane. The cells had only a few secretory granules that did not stain heavily.

IV. ESTROGEN INDUCES PITUITARY TUMORS

A. PRIMARY TUMORS

Long-term treatment with estrogen induces pituitary adenomas that are derived from the multiplication of prolactin-producing cells and cells with-

out granules (Clifton and Meyer, 1956). In the early weeks of treatment, the pituitary glands show the same morphological changes that occur with short-term treatment—hypertrophy; increases in the amount of rough endoplasmic reticulum, again in rows and whorls; and increases in the size of the Golgi apparatus. Cells are found in all stages of granulation, and mitoses occur in both granulated acidophil cells and chromophobe cells. At later stages, most cells have only a few small granules (Clifton and Meyer, 1956; Furth and Clifton, 1966; Olivier *et al.*, 1975; DeNicola *et al.*, 1978).

The ability of estrogen to cause these pituitary tumors varies with the strain of rat. Dunning and co-workers (1947) noticed that the Fischer 344 strain of rat was susceptible to pituitary tumor induction. Stone and associates (1979) found that estrogen did not increase pituitary weight in Sprague Dawley rats after 130 days of treatment, but increased the weight of the pituitary glands in ACI rats to 2–7 times as much as in control glands after 130 days. The uteri of the ACI rats were not larger, but those of the Sprague Dawley were, after estrogen treatment. Wiklund *et al.* (1981a) compared pituitary and uterine growth in ovariectomized rats using the Fischer 344 strain, in which tumors develop after 2 months of estrogen treatment, and in the Holtzman strain, in which there is no tumor development within this time. They found that estrogen rapidly increased uterine wet weight over threefold in both strains, and this increase reached a plateau after 2 weeks. Estrogen increased prolactin synthesis, expressed as a percentage of total protein synthesis in the gland, from 7% to about 40% in both strains. Estrogen, however, increased pituitary wet weight six- to tenfold in the Fischer rats, but caused no change in the Holtzman rats. These experiments indicate that the uterus in the Holtzman rats can respond to estrogen by growing and that the pituitary gland in the Holtzman rats can respond to estrogen in other ways, even though it does not grow, so the lesion is not just a general loss of responsiveness. Wiklund *et al.* (1981a) showed that pituitary glands transplanted under the kidney capsule of an F_1 recipient have the same tendencies to develop tumors as they do *in situ*, indicating that the lesion is actually in the pituitary gland itself. A series of crosses between Holtzman and Fischer parents and offspring indicate that a small number of loci, but more than one, are probably involved (Wiklund *et al.*, 1981b). These studies were continued by Wiklund and Gorski in 1982, when they demonstrated that estradiol and diethylstilbestrol both increase the weight of the gland and the amount of DNA synthesis, but DNA synthesis does not remain elevated in Holtzman rats. Estrogen increases DNA synthesis in both strains by 2 days, but synthesis returns to untreated levels after 1 week of treatment in the Holtzman strain and remains elevated in the Fischer strain. The increase in the weight of the gland is also maintained in the Fischer rats, but not in the Holtzman rats. The Holtzman strain, therefore, seems to have a mechanism to shut off estrogen-stimulated growth.

B. TRANSPLANTABLE TUMORS AND CELL LINES

Pituitary tumors that have been induced by estrogen can be transplanted to other animals. Exogenous estrogen is usually required at first for growth of these transplanted tumors, but the tumors will grow autonomously with continued passages (reviewed in Furth and Clifton, 1966). Clifton and Furth (1961), in one of the more thorough studies of effects of various doses of estrogen, found that in the early passages transplanted tumors appear to become progressively more sensitive to diethylstilbestrol. The response of the early passage tumors was biphasic; doses of diethylstilbestrol that maximally stimulated tumor growth had little effect on the weight of the normal gland, and higher doses that caused no stimulation of tumor growth compared to controls did stimulate the normal gland to grow. Estradiol stimulated growth of normal glands with the same dose-response pattern, whether they were in their usual position or transplanted under the kidney capsule. Because connections to the hypothalamus were not necessary to see the difference in sensitivity between normal glands and tumors, the difference could not be explained by hypothalamic factors influencing the normal gland. These data suggest that at least some tumors may no longer require exogenous estradiol, because they have become sensitive enough to respond to endogenous estrogens.

There are two autonomous prolactin-producing tumors, MtTF₄ and 7315a, that were each initiated by estrogen but whose growth is now strongly inhibited by high doses of estrogen (10 µg of 17β-estradiol daily) (Morel *et al.*, 1982; Lamberts *et al.*, 1982). Lamberts and co-workers found that growth of the pituitary tumor was inhibited by estradiol, but the uterus and the pituitary gland both showed a normal weight gain in response to estradiol, indicating that the inhibitory effect of estradiol is specific for the tumors. At present, the inhibitory effect has only been seen in animals and is not understood; it is of interest in part because large doses of estradiol are used to treat other tumors thought to be estrogen dependent, such as breast carcinomas (Hellman *et al.*, 1982).

Some of these autonomous hormone-producing tumors have been used to establish cell lines that grow continuously in culture. One of these, the GH cell strain, was started from a tumor induced by irradiating the head of a female rat (Tashjian *et al.*, 1968; Takemoto *et al.*, 1962). This tumor, MtTW5, grew in female rats after transplantation without needing exogenous estrogens. Two laboratories have found that injections of cell lines established from this tumor will give rise to tumors only in females and not in males (Sorrentino *et al.*, 1976; Sonnenschein *et al.*, 1973). The difference was interpreted as a requirement for estrogen, because both laboratories found that treating males with estrogen enabled the injected cells to form

tumors. Sorrentino and co-workers, however, measured estradiol levels in the serum of the rats and found that females, with 25 tumors in 25 animals, had serum estradiol levels of 40 pg/ml, and males, with no tumors in 14 animals, had levels of 20 pg/ml. Treatment with exogenous estradiol raised serum levels to over 500 pg/ml. Ovariectomized females had estradiol levels of less than 12 pg/ml, but 15 of 24 animals developed tumors, although the average tumor mass was much smaller than in normal females. Therefore, although it is clear from the two papers that large doses of estrogen stimulate the growth of these tumors in both sexes, it seems unlikely, considering the tumors in the ovariectomized females, that the twofold difference in serum estradiol levels accounts for the all-or-none effect in females versus males. Other differences in males and females besides average serum estradiol levels must be important.

Sorrentino and co-workers did not see stimulation of growth of the cells in tissue culture and concluded that estrogen acts indirectly, stimulating other organs to make growth factors. Estrogen effects in tissue culture, however, have been difficult to see reproducibly. One reason for the difficulty is the presence of endogenous estrogens in the serum usually used to supplement the medium in which the cells are grown. Estradiol can stimulate cell growth in cultures at concentrations as low as 10^{-11} M (Lippman *et al.*, 1976), and the amount of estrogen present in the serum usually used to supplement medium may be higher by one to three orders of magnitude—there is wide variation among serum lots. In order to remove endogenous estrogens, serum has been treated with dextran-coated charcoal. This procedure obviously removes many factors besides estrogens that may be necessary to see all the estrogen effects. In addition, estrone sulfate, not usually measured in assays of estradiol, is not extracted from serum by charcoal treatment and can act as a precursor for more active estrogens (Vignon *et al.*, 1980). These problems may explain why charcoal-treated serum has been difficult to use to measure reproducible estrogen effects in breast or pituitary cultures.

A second approach has been to develop a serum-free medium in which all of the components are known. So far, using GH cells in serum-free medium, laboratories have seen no effect of estrogen on cell growth or a decrease in cell growth at physiological concentrations (Hayashi and Sato, 1976; Tixier-Vidal *et al.*, 1983; Brunet *et al.*, 1982). Wyche and Noteboom (1979) also reported that estradiol inhibited growth of a human pituitary cell line in defined medium, but this inhibition occurred at concentrations above $1 \mu\text{M}$, concentrations higher than those found physiologically. A medium that is satisfactory for expressing one effect of estrogen may not be sufficient to allow expression of other effects. For example, Evans *et al.* (1981) found that induction of ovalbumin gene transcription by estrogen required a somatomedinlike activity, but induction of the conalbumin gene did not. Serum-free

medium that has been developed for the expression of one effect may not allow the expression of all.

We have reduced the amount of endogenous estrogens in medium by using serum from castrated horses, or geldings. The main source of estrogen in the male is the conversion of testosterone from the testis to estrogens in the peripheral tissues, such as skin and fat. Gelding serum, therefore, has low levels of estrogen, but other compounds have not been removed. Using serum from one horse, we have found maximum stimulation occurring between 1 and 3×10^{-11} M estradiol; the doubling time was decreased from roughly 2.5 to 1.5 days (Amara and Dannies, 1983). This response was biphasic; the rate of growth decreased between 3×10^{-11} and 3×10^{-9} M. We know that these effects of estradiol depend on factors present in the serum for two reasons. First, the increase in growth was substantially less after the serum had been heat-inactivated, and, second, the effect was much smaller in a batch of commercial gelding serum that we tried. Although we have found that the estrogen stimulation of growth has been reproducible with serum from the same horse, it appears to be necessary to screen batches of serum to use gelding serum for estrogen stimulation.

It would be satisfying to be able to relate the biphasic response we see with low concentrations of estradiol on GH cell growth to the biphasic effect of diethylstilbestrol on tumor growth at concentrations too low to affect normal pituitary growth (Clifton and Furth, 1961). Both the increase and the decrease in the rate of growth in GH cells fell within concentration ranges that could be mediated by the estrogen receptor characterized in these cells (discussed in Section VIII). The dose-response relationship of estrogen concentrations to the ability of GH cells to form tumors in animals is not known, and the response may not be biphasic. Stimulation of growth in serum is different from the inhibition seen by Tixier-Vidal and co-workers in serum-free medium, and it is not clear which conditions are more relevant to growth in animals. Estrogen obviously affects many processes in prolactin-producing cells, and changes in some of these properties might enable the cells to grow differently on plastic without changing the ability to grow as a tumor in animals. Furthermore, estrogen has been shown to increase blood supply to the pituitary gland (Elias and Weiner, 1983), an effect that could be a cause of increased tumor growth that would not show up in tissue culture. One method to determine whether the estrogen response in culture is related to effects in animals is to isolate stable variants from an estrogen-sensitive population in culture and ascertain whether the response of tumor growth in animals also changes. These types of experiments have not yet been done, either for GH cells or for breast cell lines, in which the stimulatory effects of estrogen on growth in monolayer culture have been more extensively studied and such variants are available.

V. ESTROGEN STIMULATES SYNTHESIS OF PROLACTIN

Many of the early studies, some of which have been cited here, showed that estrogen increases serum prolactin levels. Meites and his co-workers demonstrated that this increase is a direct effect on the pituitary gland by showing that incubating glands in culture for days or for 12 hours with estradiol caused more prolactin to accumulate in the medium (Nicoll and Meites, 1962; Lu *et al.*, 1971).

An increase in prolactin in the medium over a period of hours or days may reflect many processes—release of previously synthesized stores, increased synthesis of prolactin, increase in the number of prolactin-producing cells, or stabilization of intracellular prolactin. Later work showed that treating rats with estradiol *in vivo* increased the amount of [^3H]leucine incorporated into prolactin by the glands *in vitro* (MacLeod *et al.*, 1969; Maurer and Gorski, 1977) and that incubating dispersed cells with estradiol also caused the same increase (Lieberman *et al.*, 1978). A control to show that incorporation was linear over the time period measured seemed routine, but turned out to be important. Incorporation of leucine over a period of hours can reflect a combination of synthesis and breakdown, and changes in incorporation can reflect changes in either process. If incorporation of the radioactive amino acid is linear over the incubation period, then the rate of synthesis alone is the primary effect measured (Schimke and Doyle, 1970). More recent studies on prolactin degradation have indicated that [^3H]prolactin can be rapidly broken down before it is secreted from the cells, both in pituitary halves or in monolayer cultures, and that the amount of degradation may vary from one experiment to another (Shenai and Wallis, 1979; Maurer, 1980). These reports on degradation emphasize the need for caution in interpreting changes in leucine incorporation as changes in synthesis.

As techniques for quantitating mRNA were developed, these studies on prolactin synthesis were extended to show that estrogen treatment increased prolactin mRNA, measured by activity in a cell-free translational system or by quantitating the sequences by hybridization (Stone *et al.*, 1977; Ryan *et al.*, 1979; Seo *et al.*, 1979a). The studies of Gorski and co-workers showed that the increase in message activity and in messenger sequences agreed well with the increase in prolactin synthesis when male rats were treated for 3 or 6 days with estrogen (Ryan *et al.*, 1979) and that, in dispersed cell cultures from male rats, the increase in prolactin synthesis corresponded to the increase in prolactin mRNA (Lieberman *et al.*, 1982). Seo *et al.* (1979a), however, found that estrogen treatment of male rats for 1 week caused a 3-fold increase in prolactin synthesis, but an 11-fold increase in prolactin message activity; at longer times, the two parameters agreed. In a second

paper, Seo *et al.* (1979b) again demonstrated discrepancies between increases in prolactin synthesis and increases in prolactin message. They found that single injections in male rats increased message activity and sequences without measurable changes in prolactin synthesis. Chronic treatment, however, resulted in equal increases in prolactin message levels and synthesis. This group had shown that [^3H]leucine incorporation was linear over the time period used and, therefore, they did not appear to be underestimating the amount of synthesis because of degradation. The experiments of Gorski and co-workers, who saw agreement, and Seo and co-workers, who did not, used different species of rats, and there are other technical differences that may explain why the different results occurred. It appears that estrogen, under certain conditions, can induce prolactin message in male rats faster than it can induce the capacity to translate this message. There is one reservation about this conclusion; estrogen-induced increases in prolactin synthesis and production have not been seen reproducibly in culture using medium containing charcoal-treated serum or serum-free medium (Raymond *et al.*, 1978; West and Dannies, 1980; Maurer, 1982a; Giguère *et al.*, 1982), and the reasons for this variation have not been determined. It may be that it is easier to see increases in mRNA than in prolactin synthesis *in vitro* for reasons unrelated to physiological changes.

Increases in prolactin mRNA levels may reflect stabilization of existing message or synthesis of new message. By giving estrogen to ovariectomized female rats, isolating the nuclei, and measuring incorporation of [^{32}P]UTP into prolactin messenger RNA precursors, Maurer (1982b) determined that estrogen does increase synthesis of new message. Estradiol increased incorporation within 20 minutes of injection into the animal, indicating that it has a rapid effect on induction of transcription of the prolactin gene.

A question that has been raised is whether estrogen increases the synthesis of prolactin by increasing the number of cells synthesizing prolactin. Experiments with DNA synthesis inhibitors, which have been performed, have to be subject to reservations because of the toxicity of the drugs. Lieberman *et al.* (1982) showed that the increase in prolactin synthesis is greater than the increase in prolactin cell number, which suggests that the increase in synthesis cannot be accounted for by the change in cell number. Prolactin cells, however, are heterogeneous, and different morphological populations are not affected in the same way by physiological changes that affect estradiol serum levels (Merchant, 1974; Sato, 1980). Therefore, one could postulate a preferential increase in cells highly active in synthesizing prolactin to account for the discrepancy between cell number and synthesis. The best evidence for a direct effect of estrogen on the prolactin gene itself is the rapid effect on transcription of prolactin RNA sequences; an effect occurring in 20 minutes cannot be accounted for by cell division.

Estrogen can stimulate prolactin synthesis in tumor cell lines as well as in primary cultures (Tashjian and Hoyt, 1972; Haug and Gautvik, 1976; Herbert *et al.*, 1978; Tixier-Vidal *et al.*, 1983). Under conditions where we found both an increase in growth and an increase in prolactin synthesis, we have examined a series of estrogenic compounds and found that growth of GH cells was always tenfold more sensitive to estrogen stimulation than was prolactin production (Amara and Dannies, 1983, and unpublished data). Such differences in the sensitivity of various responses to steroids have been seen with glucocorticoids in other cell lines (Mercier *et al.*, 1983). At present, it is not known whether the difference in sensitivity of the two responses is also seen at the level of mRNA transcription and accumulation or whether it is caused by posttranscriptional factors. With the techniques now available, it will be possible to determine this.

VI. ESTROGEN INCREASES PROLACTIN STORAGE

We began studies in GH cells to attempt to cause these cells to differentiate into a more normal phenotype. One of the noticeable differences between GH cells and normal cells is the amount of prolactin they contain; GH cells store a much smaller proportion of the prolactin that they secrete over a 24-hour period than normal prolactin cells do, and they lack the dense secretory granules seen in electron micrographs of normal prolactin cells (Tixier-Vidal, 1975). We found that treating the cells with insulin and estradiol caused an increase in intracellular prolactin that was larger than the increase in prolactin in the medium (Kiino and Danies, 1981, 1982). In early experiments, we found three- to sevenfold increases in intracellular prolactin after treating the cells for 6 days with 180 nM insulin and 1 nM estradiol, compared to increases of less than twofold in the amount of prolactin produced in the medium. Insulin and estrogen acted synergistically to increase intracellular prolactin, but were less than additive in increasing extracellular prolactin. In subsequent experiments, we have shown that the effect was larger when we plated cells at lower initial densities or added 10 nM epidermal growth factor. The three hormones together increased intracellular prolactin 10- to 30-fold, while increasing extracellular prolactin 2- to 4-fold. These experiments were performed in medium containing serum that had not been treated in any special way to remove exogenous steroids, so this effect was apparently less sensitive to estradiol concentrations than the stimulation of growth was.

We have shown that prolactin synthesis was specifically increased by the combination of insulin and estrogen, but that the increase was not large enough to account for the increase in intracellular prolactin (Kiino and Dan-

nies, 1982). We have shown also that the increase in the time it takes newly synthesized, rapidly released prolactin to be discharged from the cells in large quantities (about 1 hour in GH cells) was not affected by this treatment. Although intracellular prolactin content increased as cells passed through the cell cycle, insulin and estrogen increased prolactin storage at each stage of the cycle and did not affect the cell cycle distribution (Kiino *et al.*, 1982).

There appear to be at least two components of prolactin release in the pituitary cells (reviewed in Dannies, 1982), and how these relate to the morphology has not been clearly demonstrated. The most likely form for stored prolactin, based on current knowledge, is secretory granules. Using another clonal cell strain, 2B8, McGill (1983) showed that insulin and estrogen increased the percentage of the area of the cell occupied by granules sixfold and that these granules contained prolactin, visualized by immunostaining. In 2B8 cells, as in GH cells, granules are not uniformly distributed among the cells, so this kind of study has to be reported by the quantitative examination that McGill did.

We began these experiments to see if we could cause differentiation of GH cells into more normal cells, but the ability to induce storage is not restricted to continuous cell lines. We have found more recently that insulin and estrogen also increased intracellular prolactin stores in primary cultures of normal male rat pituitary cells in gelding serum (C. M. Van Itallie and P. S. Dannies, unpublished results). There are some suggestions that such processes also occur in intact animals; estrogen increased the pituitary content of prolactin (Meites *et al.*, 1972; Lloyd *et al.*, 1973) as well as the amount of prolactin that could be released (Labrie *et al.*, 1980). In the morphological studies of Osamura *et al.* (1982), however, castration resulted in the disappearance of large secretory granules and the appearance of many small secretory granules; these disappeared after injection of estradiol, a result that does not appear to agree with the studies performed in culture. The situation in animals is complicated by interactions of estrogen with the hypothalamus as well as with the pituitary gland, interactions that may make it more difficult to distinguish effects on storage from other effects. The ability to manipulate storage conditions in cells in culture provides a way of resolving the proteins involved in creating storage. Once these are identified, the question of whether they also change in intact animals can be directly answered.

A role of estrogen in increasing the storage of prolactin by decreasing the ability of the cells to degrade prolactin has been suggested. After prolactin secretion in lactating rats is prevented by removing the suckling pups, the contents of secretory granules can be seen in the lysosomes (Smith and Farquhar, 1966). As mentioned previously, there is biochemical evidence

that pituitary glands and cells will degrade newly synthesized prolactin under some conditions. Dopaminergic agents enhance this ability (Maurer, 1980; Dannies and Rudnick, 1980). Nansel *et al.* (1981a) found that dopamine increased the activity of β -glucuronidase, an enzyme found in lysosomes and microsomes of the cells. They reported that the levels of β -glucuronidase decreased as estrogen levels increased and that estradiol decreased enzyme levels in ovariectomized rats (Nansel *et al.*, 1981b). Other laboratories, however, did not see similar changes. Nagy and co-workers (1983) did not see changes in acid phosphatase, another enzyme found in the lysosomes, or in β -glucuronidase after dopamine or estrogen treatment, and Chowdhury and Sarkar (1983) found a decrease in β -glucuronidase activity, but an increase in acid phosphatase activity, with estrogen treatment. Interpretation of any experiment is obviously complicated by the heterogeneous populations of cells in the gland; it is not clear in which cells changes occur when differences are found. The only conclusion that can be reached at this time is that changes in enzyme activity sufficiently large to be detectable are not necessary for the effects on prolactin production. Estrogen regulation of prolactin degradation remains an unproven possibility. Regulation of intracellular prolactin stability, if it exists, may not occur only at the lysosomal level; evidence in several systems suggests a nonlysosomal component of the degradation of secretory proteins (Bienkowski, 1983).

VII. ESTROGEN CHANGES THE RESPONSE OF PROLACTIN CELLS TO AGENTS THAT AFFECT PROLACTIN RELEASE

In addition to causing direct increases in prolactin synthesis and storage, estrogen affects the ability of other factors to stimulate or inhibit prolactin release. This effect was first shown for thyrotropin-releasing hormone (TRH) in intact animals. DeLéan and co-workers (1977a) showed that estrogen injections increased the amount of prolactin released in response to TRH. This effect appeared to be physiologically relevant, because prolactin responses to TRH were lowest in diestrus and highest in proestrus, and TRH binding to the pituitary gland was lowest in diestrus and highest in proestrus (DeLéan *et al.*, 1977b). The ability of estrogen to increase the binding of TRH in GH cells and to directly increase TRH-induced prolactin release in normal cells and GH cells has been demonstrated (Gershengorn *et al.*, 1979; Giguère *et al.*, 1982).

At the same time that estradiol enhanced the effects of TRH, it lessened the inhibitory action of dopaminergic agents (Raymond *et al.*, 1978). We extended these studies to show that estradiol caused a shift in the half-

maximally effective concentration of dihydroergocryptine, a dopaminergic agonist, that ranged from 10- to 30-fold and, in addition, reduced the maximal amount of inhibition obtained to about half the amount of inhibition in untreated cultures (West and Dannies, 1980). These effects of estradiol occurred within 2 days, a time span that could be relevant to the estrous cycle, and occurred with a maximal effect at 50 pM estradiol. Measurements of plasma estradiol concentrations during the diestrous period of the rat range from 7 to 370 pM (Butcher *et al.*, 1974; Brown-Grant *et al.*, 1970; Shaikh and Shaikh, 1975), so these effects of estradiol may have physiological relevance.

Although Heiman and Ben-Jonathan (1982) showed that dopaminergic receptors varied during the estrous cycle, DiPaolo *et al.* (1979) found no effect of estradiol on pituitary dopaminergic receptors when the steroid was injected into rats, and suggested that a postreceptor step was involved. Giguère *et al.* (1982) have found that estrogen increased the amount of release occurring in response to isobutylmethylxanthine, a phosphodiesterase inhibitor. It is difficult to interpret this result cleanly, even if the only effect of isobutylmethylxanthine is to inhibit phosphodiesterase activity, except to say that a receptor is not involved. Estrogen may have increased the activity of adenylate cyclase so that more cyclic AMP can accumulate when enzymatic breakdown is prevented, and therefore more release occurs. Alternatively, estrogen may have affected some step other than cAMP formation, such as storage, to make the release of more prolactin possible. The estrogen-induced changes in response to TRH and dopamine may be completely accounted for by changes in receptor number and increases in intracellular stores of prolactin, but further investigations may show that other mechanisms are involved as well.

In serum-free medium, almost any component that we added enhanced or inhibited the ability of estrogen to decrease the response to dopamine (B. West and P. S. Dannies, unpublished results). One problem is to determine which of these effects occur *in vivo*; effects found in culture may be different from those in the animal. For example, Maurer (1980) found that estradiol could not overcome thyroid hormone inhibition of prolactin synthesis in primary cultures, but DeLéan *et al.* (1977a) found that estrogen could overcome thyroid hormone inhibition of prolactin production in intact animals. It is possible now to determine in the animal whether the effect of estrogen observed by DeLéan *et al.* is an effect on synthesis and whether the *in vivo* results are different from the results obtained in tissue culture.

VIII. RAPID EFFECTS OF ESTROGEN

Most, if not all, of the estrogenic effects described up to this point are probably mediated through the same kinds of estrogen-receptor genomic

interactions that occur in other systems. Pituitary glands and GH cells have estradiol receptors similar to those that have been characterized in other systems, with reported affinities ranging from 2×10^{-10} to 1×10^{-9} M (Eisenfeld, 1970; Leavitt *et al.*, 1969; Vreeburg *et al.*, 1975; Haug *et al.*, 1978; Roos *et al.*, 1980; Mester *et al.*, 1973). The best evidence that these binding sites mediate biological responses comes by analogy with results obtained using receptor mutants in other systems. Cells with defective steroid binding show altered responses to steroids (reviewed by Strobl and Thompson, 1984). It seems reasonable to assume that, if pituitary cells with altered estrogen receptors were examined, many or all of the responses described would be affected. There may, however, be nongenomic membrane-mediated events. Estradiol at 1-nM concentrations caused a rapid increase (within 1–2 minutes) in formation of action potentials in 30% of the cells in a clone of GH cells (Duffy *et al.*, 1979). These action potentials may have a calcium component because they could be blocked by D600, a calcium channel blocker. 17α -Estradiol was not as effective in stimulating action potentials, but blocked the action of 17β -estradiol. These effects were so rapid that it seems likely that they were not mediated by genomic interactions. Zysek *et al.* (1981) have found that 17β -estradiol concentrations of 10^{-8} M or higher stimulated release of prolactin 10 minutes after the steroid was added to the same clone of GH cells in which the electrophysiological effects were found. At times longer than 10 minutes, the large amount of basal release obscured the stimulation. Because increasing intracellular calcium concentrations can cause prolactin release, it is possible that the rapid increase in formation of action potentials accounts for estrogen-stimulated release. There are rapid effects of steroids seen in other systems, reviewed by Baulieu (1978) and Duval *et al.* (1983). These effects include rapid changes in the firing rates of neurons at steroid concentrations that sometimes are 1 nM or less. One of the ways in which prolactin cells resemble nerve cells is in their ability to show action potentials. Both cell types apparently have the ability to respond to steroids with rapid membrane effects.

IX. EFFECTS OF ANTIESTROGENS

The reports of the effects of estrogen antagonists on prolactin levels in intact normal animals vary. Clomiphene has been found to decrease basal serum prolactin levels (Burdman *et al.*, 1979) and to cause a twofold increase (Tobias *et al.*, 1981). Tamoxifen had no effect on basal serum prolactin in studies by Nicholson and Golder (1975), Spona and co-workers (1980), and Leibl and workers (1981). On the other hand, Spona and Leibl (1981) and Prysor-Jones and co-workers (1983) found increases in serum prolactin levels

after tamoxifen. Nagy *et al.* (1980) found no change in serum prolactin levels in Buffalo rats, but a threefold increase in Wistar Furth rats, although this change was evidently not statistically significant. Estrogen antagonists are known to be partial agonists, and it is therefore not necessarily surprising that there are conditions under which they show stimulatory activity. Agents that have only partial agonist activity should partially block the activity of agents with full activity. Jordan *et al.* (1975), Jordan and Koerner (1976), and Clemens *et al.* (1983) found that tamoxifen could partially block the estrogen-induced rise in serum prolactin, but Spona *et al.* (1980) and Leibl *et al.* (1981) found that tamoxifen increased the estrogen-induced rise in serum prolactin levels in the same experiments where it had no effects on its own.

Tamoxifen effects on prolactin synthesis have been measured by treating rats with the compound and then removing the pituitary glands and incubating them with [^3H]leucine. Several laboratories have observed decreased incorporation of leucine into prolactin, either of basal levels or of estrogen-induced increases (Spona and Leibl, 1981; Nagy *et al.*, 1980; DeQuijada *et al.*, 1980). The decrease was seen even in cases when serum prolactin levels or prolactin levels in the cultures increased or were not affected. The interpretation of these results as decreases in synthesis must be done with caution, because the incubation times were as long as 4 hours. In view of the previously described occurrence of degradation, it is probably necessary to confirm under each set of experimental conditions that changes in synthesis are being measured. Spona and Leibl (1981) found that tamoxifen had no effect on basal incorporation of [^3H]leucine into prolactin, but caused increases in translatable prolactin mRNA, and tamoxifen decreased estrogen stimulation by both techniques. If the leucine incorporation is a good measure of synthesis, then the discrepancy between prolactin mRNA activity and leucine incorporation may indicate control of a translational process in the gland.

The results of estrogen antagonists on tumor growth also vary. The growth of pituitary tumor 7315a was inhibited by tamoxifen and was also inhibited by estrogen as previously discussed (DeQuijada *et al.*, 1980; Nagy *et al.*, 1980; Lamberts *et al.*, 1981). Nagy *et al.* (1980) found that the growth of MtTW15 tumors was inhibited by tamoxifen, but Winneker *et al.* (1981) found that the growth of these tumors was stimulated by tamoxifen and other antiestrogens.

Many factors may account for the differences summarized here. Results in intact animals obviously can occur through indirect as well as direct effects on the gland. The variability in experiments in intact animals may be caused by the age or sex of the animal, whether the animal has been castrated or ovariectomized, the dose of antagonist administered, the times of administration, and perhaps even the strain of rat; more than one of these factors

may affect the response. The studies cited above have too many differences among them to be able to determine causes. It is important to resolve eventually what conditions cause estrogen antagonists to elevate prolactin levels in the serum. Clomiphene is used clinically as a fertility drug, but hyperprolactinemia causes infertility. If clomiphene elevates prolactin levels in some patients, the effect could interfere with the desired stimulation of fertility. Estrogen antagonists are also used to treat breast cancer. Again, if prolactin is elevated in some patients by these agents, it may interfere with inhibition of growth of the cancer.

The actions of estrogen antagonists have been widely studied in other tissues; one review of this work is that by Sutherland and Murphy (1982). Estrogen antagonists compete with estrogen for binding to the estrogen receptor. Unoccupied estrogen receptors are primarily found in the cytoplasm when cells are homogenized; binding of estrogen or most estrogen antagonists causes the receptor to be retained in the nucleus. The retention of the estrogen receptor in the nucleus is longer with most estrogen antagonists than it is with estrogen. Tamoxifen appears to behave in the pituitary gland as it does in other systems, causing prolonged retention of the estrogen receptor in the nucleus (Bowman *et al.*, 1982; Kirchoff *et al.*, 1983). Sutherland and Murphy point out in their review that tamoxifen has direct effects on estrogen biosynthesis, prostaglandin synthesis, and enzymes of carbohydrate metabolism; these effects are not always reversed by estrogen. Winneker *et al.* (1983) have shown that tamoxifen can bind to a site on the serum low-density lipoprotein, an effect that may interfere with cholesterol synthesis. There is also an intracellular binding site for estrogen antagonists distinct from its binding site on the estrogen receptor. Studies with variant cell lines resistant to tamoxifen suggest that this binding site is important for the action of these compounds in inhibiting the growth of a breast cancer cell line (Faye *et al.*, 1983). Although estrogen can compete with the anti-estrogens to prevent the growth inhibition, Faye and co-workers suggest that this competition may be at a physiological level, rather than at the level of the receptor. Therefore, the various results reported with estrogen antagonists in animals may not all be mediated through the estrogen receptor—in fact, it is not yet proven that any effects are.

Lieberman *et al.* (1983a) have approached this question pharmacologically by using cell cultures of normal pituitary glands, which avoids actions at sites other than the gland itself. They have measured estrogen-induced prolactin synthesis and its inhibition by antiestrogens. If an antagonist is competitive, it ought to shift the dose-response curve of the agonist without affecting the shape of the curve. Increasing concentrations of antiestrogens ideally should result in a parallel series of estrogen dose-response curves, with increases in the apparent half-maximally effective concentration of estrogen. Interactions

in a whole cell are never ideal, but Lieberman *et al.* (1983a) did show at least that two concentrations of hydroxytamoxifen increased the apparent half-maximal concentrations of estradiol. Therefore, the antagonist appears to behave in a competitive manner. Lieberman *et al.* (1983b) present a second kind of pharmacological evidence that antiestrogens exert their effects through the estrogen receptor; they showed that the relative potencies of the compounds in inhibiting prolactin synthesis were similar to the relative binding affinity for the estrogen receptor. At present, pharmacological correlations are the best evidence for the action of antiestrogens. Cell lines with mutations in estrogen receptors and antiestrogen binding sites are not yet available for pituitary cell lines.

In the series of experiments using cells in culture, described above, estrogen antagonists behaved as antagonists are expected to behave, suggesting that culture systems may be simpler than intact animals. This is not always the case. In GH cells grown in normal horse serum and fetal calf serum, we showed that several estrogen antagonists stimulated prolactin production after a week or more of treatment, but estradiol did not (Dannies *et al.*, 1977). Estrogen actually prevented the increase in prolactin production. The lack of effect of estradiol may have been caused by endogenous estrogens in the serum, since in this series of experiments we made no effort to remove them, but because the estrogen effect may depend on other serum factors, this explanation is not the only one possible. If the antiestrogens were increasing prolactin production by acting as estrogens, then they appeared to be better estrogens than estradiol under those circumstances. More recently, we have used antiestrogens in medium containing gelding serum, and we found that most antiestrogens were partial agonists in stimulating both prolactin production and cell growth (J. F. Amara and P. S. Dannies, unpublished results). Whether the difference in the results in GH cells and those in normal cells is caused by trivial differences in culture conditions or by more interesting differences in gene induction in normal and GH cells is not yet known.

X. CONCLUSION

Estrogen affects prolactin cells in several ways. With the technology now available, it will be possible to use pituitary glands and cell lines to learn more about how estradiol regulates transcription. It is also now possible to isolate messages induced by estrogen in addition to prolactin. These other induced proteins must regulate growth and secretion, and by learning more

about these proteins we will know more about how these processes function and are regulated, both in cell culture and in animals.

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